

Statistical Mechanics: Boltzmann Weights, Partition Functions, and the Ideal Gas

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Chapter 1

Boltzmann Weights, Partition Functions, and the Ideal Gas

Overview. We derive the equilibrium probability distribution by maximizing entropy at fixed normalization and mean energy. The partition function then yields the mean energy, entropy, and temperature by differentiation. The ideal gas supplies the first full example, followed by a gravitational extension and a preview of energy fluctuations.

1.1 The Constrained Entropy Problem

Begin with a large collection of $\mathcal{N}$ identical, very weakly interacting replicas of the same subsystem. If n_i replicas occupy the i -th state, then

$$p_i = \frac{n_i}{\mathcal{N}}, \quad \sum_i p_i = 1. \quad (1.1)$$

The multinomial counting of the preceding lecture gives the entropy per replica,

$$S[p] = - \sum_i p_i \log p_i. \quad (1.2)$$

It is worth viewing this as an ordinary function of all the variables $p_1, p_2, \dots$. Its simple additive form is what makes the calculation possible.

The replica collection has a fixed total energy $\mathcal{N}E$, where E is the mean energy per replica. The second constraint is therefore

$$\sum_i p_i E_i = E. \quad (1.3)$$

The levels E_i belong to the subsystem; the number E on the right-hand side is their probability-weighted mean. Maximizing (1.2) subject to (1.1) and (1.3) selects the overwhelmingly most probable occupation fractions.

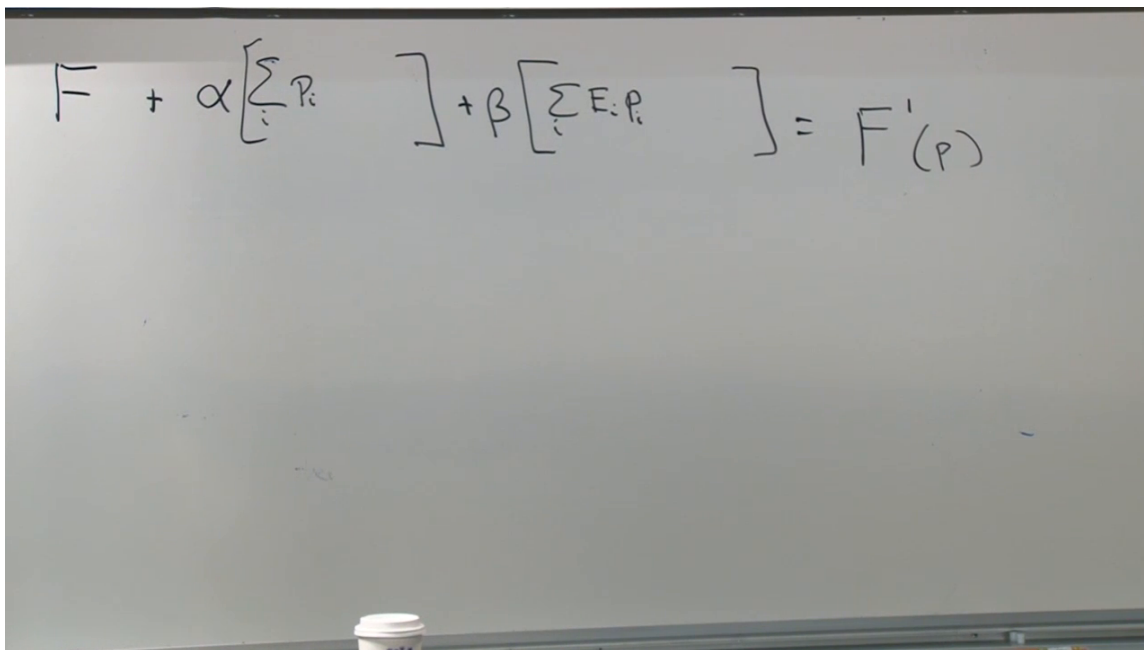


Figure 1.1: The augmented entropy functional after terms constant under variation of the probabilities have been suppressed.

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For convenience, minimize $-S = \sum_i p_i \log p_i$. Introduce one Lagrange multiplier for each constraint:

$$\begin{aligned} \Phi[p; \alpha, \beta] = & \sum_i p_i \log p_i \\ & + \alpha \left(\sum_i p_i - 1 \right) \\ & + \beta \left(\sum_i p_i E_i - E \right). \end{aligned} \tag{1.4}$$

The names α and β are not yet physical assumptions. At this stage they are simply the two numbers required to enforce the two constraints.

Question from the audience. Is the augmented function also a function of α and β ?

Response. Yes. Varying it with respect to α and β restores the two constraints. For the immediate calculation, however, we hold those multipliers fixed and vary the probabilities p_i . Their values are fixed afterward by normalization and the prescribed mean energy. ¹

When differentiating with respect to a probability, the terms $-\alpha$ and $-\beta E$ contribute nothing. This is the abbreviated functional visible on the board. They remain part of (1.4); they have merely become irrelevant to this particular derivative.

¹Lecture timestamp: 00:09:32–00:09:49.

1.2 The Boltzmann Distribution

Choose any one probability. The lecture chooses p_7 , partly because 1 looks too special and 7 is a lucky number. Nothing depends on that choice. The only calculus required is

$$\frac{d}{dp_i}(p_i \log p_i) = \log p_i + 1. \quad (1.5)$$

Thus

$$\frac{\partial \Phi}{\partial p_i} = \log p_i + 1 + \alpha + \beta E_i = 0. \quad (1.6)$$

Solving for p_i ,

$$\log p_i = -(1 + \alpha) - \beta E_i, \quad (1.7)$$

$$p_i = e^{-(1+\alpha)} e^{-\beta E_i}. \quad (1.8)$$

The first factor is independent of the state label. Write it as $1/Z$:

$$p_i = \frac{e^{-\beta E_i}}{Z}. \quad (1.9)$$

This is the first sudden payoff. A routine constrained derivative has produced an exponential law. The relative probabilities obey

$$\frac{p_i}{p_j} = e^{-\beta(E_i - E_j)}. \quad (1.10)$$

The multiplier β controls how rapidly probability falls with energy. Large positive β concentrates the distribution near the lowest levels; small positive β spreads appreciable weight farther up the spectrum. We can already see that changing β changes the mean energy, although its identification with inverse temperature still has to be proved.

1.3 Normalization and the Partition Function

Normalization determines the state-independent factor:

$$1 = \sum_i p_i = \frac{1}{Z} \sum_i e^{-\beta E_i}, \quad (1.11)$$

so

$$Z(\beta) = \sum_i e^{-\beta E_i}. \quad (1.12)$$

This is the canonical partition function. It first appeared as a normalizing constant, but that description badly understates its role. Once $Z(\beta)$ is known, thermodynamic quantities follow from its derivatives.

The energy levels in (1.12) are inputs. They come from the mechanics of the system together with fixed external conditions: the volume and shape of the box, magnetic or electric fields, and any other control parameters. Changing a control parameter changes the spectrum and therefore changes Z . For the present calculation those parameters, and hence the set $\{E_i\}$, are held fixed.

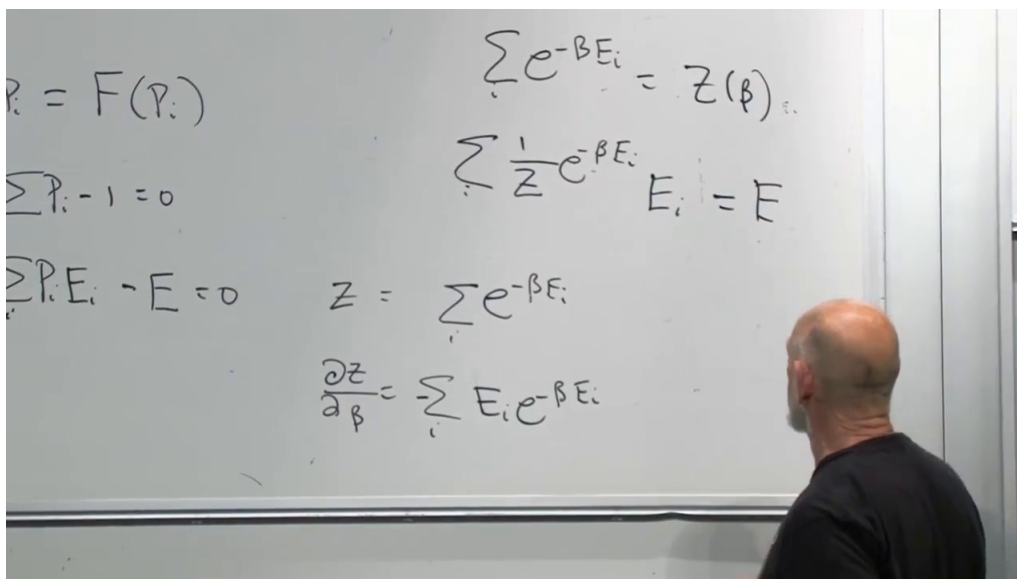


Figure 1.2: Normalization, the mean-energy constraint, and the derivative of the partition function appear together on the board.

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Question from the audience. What exactly do the axes of the exponential sketch represent? Is the vertical coordinate one particular p_i ?

Response. The plotted height is the probability assigned to a level, while the horizontal ordering may be read either as the state label i or, more physically, as the energy E_i . Every state has a fixed energy. Large β makes the probability fall steeply with that energy; near $\beta = 0$, the weights are nearly flat over the accessible levels.²

1.4 Mean Energy Hidden in the Partition Function

The second constraint determines β :

$$E = \sum_i p_i E_i = \frac{1}{Z} \sum_i E_i e^{-\beta E_i}. \quad (1.13)$$

Rather than evaluate this weighted sum afresh, differentiate (1.12):

$$\frac{dZ}{d\beta} = - \sum_i E_i e^{-\beta E_i}. \quad (1.14)$$

Comparison with (1.13) gives

$$E(\beta) = - \frac{1}{Z} \frac{dZ}{d\beta} = - \frac{d \log Z}{d\beta}. \quad (1.15)$$

The logarithm is natural because its derivative supplies the factor $1/Z$.

²Lecture timestamp: 00:19:39–00:20:09.

Question from the audience. Is the lower board equation a new equation for Z , or is it obtained from the definition above it?

Response. It is obtained by differentiating the definition $Z = \sum_i e^{-\beta E_i}$. The derivative pulls down $-E_i$. Dividing the result by Z then reproduces the weighted sum that defines the mean energy.³

Question from the audience. Does $dZ/d\beta = -EZ$ make Z an eigenfunction of differentiation?

Response. Not in the usual sense. An eigenvalue equation would have a constant multiplying Z , whereas here $E = E(\beta)$. The equation tells us the mean energy as a function of β ; it is not a fixed-eigenvalue problem.⁴

The simplicity of (1.15) is deceptive. The spectrum may be complicated, and calculating Z can be difficult, but once that single function has been found the mean energy follows by one derivative. This pattern recurs throughout statistical mechanics: elementary operations applied in the right order yield results that would have been hard to guess directly.

Question from the audience. Is E the total energy of all the replicas or the average energy per box?

Response. It is the average energy of one box. The replica collection has total energy $\mathcal{N}E$. Later, when the system itself is chosen to be a whole box containing N gas particles, E will mean the energy of that whole gas sample.⁵

Question from the audience. Can we take the logarithm of $\sum_i e^{-\beta E_i}$ term by term?

Response. No. A logarithm turns a product into a sum, not a sum into a sum of logarithms:

$$\log\left(\sum_i e^{-\beta E_i}\right) \neq \sum_i \log(e^{-\beta E_i}).$$

This distinction becomes useful later, when independent subsystems make their partition functions multiply.⁶

1.5 What Is Fixed, and What Is Allowed to Vary?

There are several layers of variables in the same formulas, and the classroom discussion deliberately separates them.

- The probabilities p_i are the original unknowns.

³Lecture timestamp: 00:25:58–00:26:37.

⁴Lecture timestamp: 00:28:49–00:29:43.

⁵Lecture timestamp: 00:30:55–00:31:10.

⁶Lecture timestamp: 00:31:10–00:31:58.

- The levels E_i are fixed once the system and its control parameters are specified.
- The multiplier α has been absorbed into the normalization $Z = e^{1+\alpha}$.
- For a prescribed mean energy E , the constraint fixes one number β . Across a family of equilibrium states, however, either $E = E(\beta)$ or $\beta = \beta(E)$ may be used.

Question from the audience. If the multipliers are numbers, why are β and E now being treated as variables?

Response. For one constrained problem they are indeed numbers. But we may compare many equilibrium problems with the same spectrum and different prescribed mean energies. Along that family, β changes with E . Fixing E determines β ; fixing β determines E . These are two readings of the same relation. ⁷

Question from the audience. Has α disappeared?

Response. Only its name has disappeared. The definition $e^{-(1+\alpha)} = 1/Z$ absorbs it into Z . Keeping both α and Z would count the same normalization parameter twice. ⁸

Energy need not be the only conserved quantity. If total momentum, particle number, angular momentum, or another quantity is constrained, its expectation value receives its own multiplier. That enlargement changes the ensemble but not the method.

Question from the audience. Does fixing only the mean energy presume too much about the system?

Response. It presumes that no other conserved quantity needs to be conditioned on in the present problem. If, for example, total momentum is also fixed, add another constraint and another Lagrange multiplier. By contrast, volume, shape, and external fields are control parameters here: they are held fixed and determine the energy levels themselves. ⁹

Question from the audience. Should α appear as another argument of the energy, and does Z depend on the levels E_i ?

Response. α is already contained in Z , so it is not an independent argument. The partition function certainly depends on the fixed spectrum as well as on β . Changing a control parameter changes the E_i , and therefore changes Z ; during the present β -derivatives the spectrum is held fixed. ¹⁰

⁷Lecture timestamp: 00:31:58–00:33:27.

⁸Lecture timestamp: 00:33:29–00:33:49.

⁹Lecture timestamp: 00:33:51–00:35:50.

¹⁰Lecture timestamp: 00:35:52–00:37:10.

Question from the audience. Are we varying the individual levels E_i , or the mean energy E ?

Response. The individual levels remain fixed. The mean energy changes because the probabilities change. Thus the family is parameterized by E or equivalently by β , while the same list of levels $\{E_i\}$ is used throughout. ¹¹

In practice temperature is often much easier to measure than the total energy of a gas. We can insert a thermometer; weighing the relativistic mass difference and invoking $E = mc^2$ would be correct but spectacularly unhelpful. To make β operational, we must now connect it to the thermodynamic definition of temperature.

1.6 Entropy Before Temperature

Entropy logically comes first. Insert the Boltzmann probabilities into (1.2):

$$\log p_i = -\beta E_i - \log Z. \quad (1.16)$$

Then

$$S = - \sum_i p_i \log p_i \quad (1.17)$$

$$= \beta \sum_i p_i E_i + \log Z \sum_i p_i \quad (1.18)$$

$$= \beta E + \log Z. \quad (1.19)$$

The normalization sum supplies the final factor of one. This is also the first hint of the Helmholtz free energy, since later $F_H = -T \log Z$ in units with $k_B = 1$.

Question from the audience. What happened to the factor $1/Z$ in the first term?

Response. It is already part of p_i . The sum $\sum_i p_i E_i$ includes the complete normalized probability and is, by definition, the mean energy E . ¹²

Question from the audience. Should there be a minus sign in $S = \beta E + \log Z$?

Response. No. The entropy begins with a minus sign, while $\log p_i = -\beta E_i - \log Z$ contains two more minus signs. They produce the two positive terms displayed in (1.19). ¹³

Temperature was defined thermodynamically by

$$dE = T dS, \quad \frac{1}{T} = \frac{dS}{dE}, \quad (1.20)$$

¹¹Lecture timestamp: 00:37:12–00:39:04.

¹²Lecture timestamp: 00:43:19–00:43:26.

¹³Lecture timestamp: 00:45:53–00:46:03.

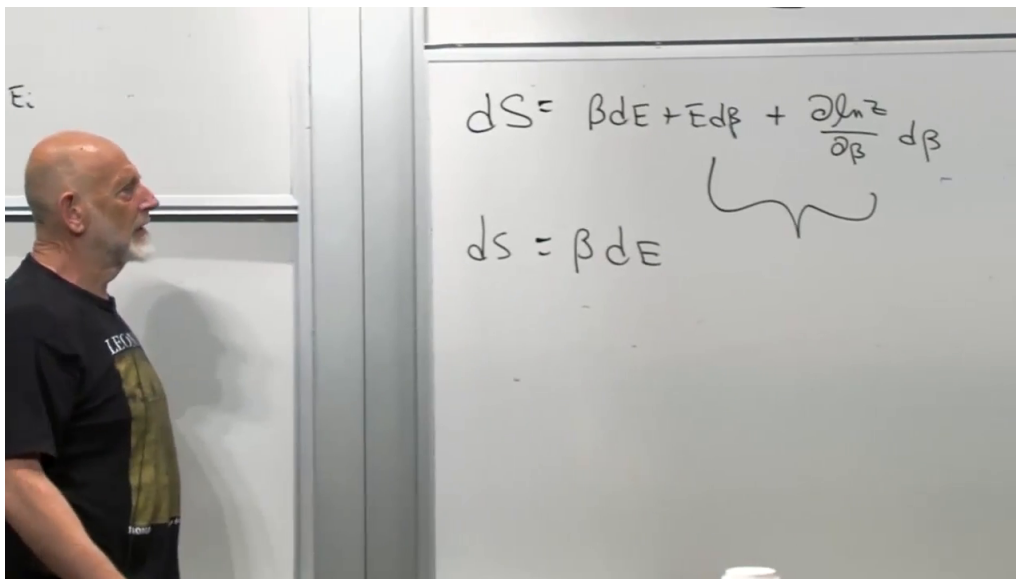


Figure 1.3: The two $d\beta$ terms cancel, leaving $dS = \beta dE$.

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with all other control parameters held fixed. A first attempt to differentiate (1.19) directly with respect to E quickly becomes cluttered because β also depends on E . Taking a total differential is cleaner:

$$dS = \beta dE + E d\beta + \frac{d \log Z}{d\beta} d\beta \quad (1.21)$$

$$= \beta dE + E d\beta - E d\beta \quad (1.22)$$

$$= \beta dE. \quad (1.23)$$

The cancellation uses (1.15). Comparing (1.23) with (1.20) gives

$$T = \frac{1}{\beta}, \quad \beta = \frac{1}{T}. \quad (1.24)$$

Question from the audience. Does this cancellation incorrectly assume that $\log Z$ is independent of energy or temperature?

Response. No. For the fixed spectrum there is one independent thermal parameter. We may write $Z = Z(\beta)$, then regard $\beta = \beta(E)$ along the equilibrium family. The total differential keeps the full $d\beta$ dependence, and the identity $d \log Z / d\beta = -E$ is exactly what makes the cancellation work. ¹⁴

Throughout most of the calculation temperature is measured in energy units, so $k_B = 1$. Restoring laboratory units,

$$\beta = \frac{1}{k_B T_K}, \quad (1.25)$$

where T_K is an absolute temperature in kelvin. It is often convenient to write $T_e = k_B T_K$ and then use $\beta = 1/T_e$.

¹⁴Lecture timestamp: 00:52:04–00:52:44.

Question from the audience. For a volume of gas, is the system one molecule or the whole collection?

Response. That depends on how the system is defined. Here the system will be the entire box of gas, so E is the energy of all its molecules. For a sufficiently dilute ideal gas, one molecule can also be treated as a subsystem because the particles do not interact; the factorization below will show the relation between the two descriptions. ¹⁵

The canonical identities can now be collected:

$$p_i = \frac{e^{-\beta E_i}}{Z}, \quad Z(\beta) = \sum_i e^{-\beta E_i}, \quad (1.26)$$

$$E = -\frac{d \log Z}{d\beta}, \quad S = \beta E + \log Z, \quad T = \frac{1}{\beta}. \quad (1.27)$$

1.7 A Classical Ideal Gas

Take N identical point particles of mass m in a box of volume V . The gas is ideal when interactions between particles can be neglected. This may be a good approximation because the forces are weak or because the gas is dilute enough that close encounters are rare.

A classical microstate is specified by all positions and momenta:

$$(\mathbf{q}_1, \dots, \mathbf{q}_N; \mathbf{p}_1, \dots, \mathbf{p}_N). \quad (1.28)$$

There are $3N$ position coordinates and $3N$ momentum coordinates, so the state is a point in a $6N$ -dimensional phase space. With no interactions, the Hamiltonian is

$$H(\mathbf{p}) = \sum_{a=1}^N \frac{\mathbf{p}_a^2}{2m} = \sum_{\mu=1}^{3N} \frac{p_\mu^2}{2m}. \quad (1.29)$$

Question from the audience. Are there $3N$ particles, and does summing over all momentum components double-count the states?

Response. There are N particles and $3N$ momentum components. The sum in (1.29) contains one quadratic term for each component. The phase-space integral then integrates each independent component exactly once, so there is no double counting from that product. ¹⁶

For a continuous classical phase space, the state sum becomes

$$Z_N = \frac{1}{N! h^{3N}} \int_{V^N} d^{3N} q \int_{\mathbb{R}^{3N}} d^{3N} p e^{-\beta H(\mathbf{p})}. \quad (1.30)$$

The lecture concentrates on the integrals and writes the same expression without the phase-space cell h^{3N} . Restoring it makes Z_N dimensionless and records the quantum origin of classical state counting. Because h is independent of β , it does not alter the energy derivative used below.

¹⁵Lecture timestamp: 00:53:32–00:54:49.

¹⁶Lecture timestamp: 01:02:29–01:04:04.

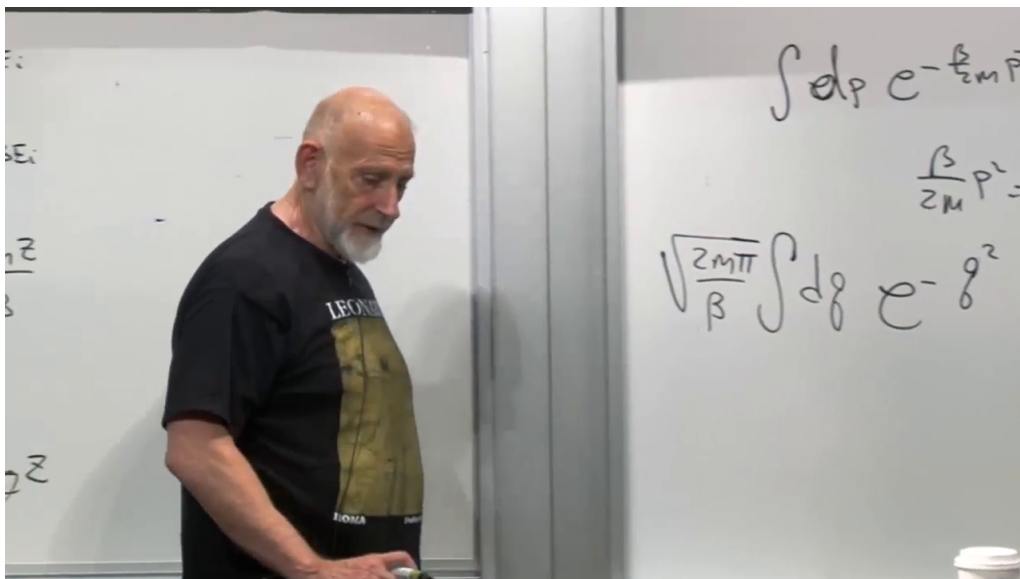


Figure 1.4: The momentum integral is reduced to the standard Gaussian by a single rescaling.

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The position integral is immediate. The Hamiltonian is independent of position, and each particle may lie anywhere in the box:

$$\int_{V^N} d^{3N}q = V^N. \quad (1.31)$$

The momentum integral factorizes into $3N$ identical one-dimensional Gaussian integrals:

$$Z_N = \frac{V^N}{N! h^{3N}} \left[\int_{-\infty}^{\infty} dp e^{-\beta p^2/(2m)} \right]^{3N}. \quad (1.32)$$

The factor $1/N!$ identifies configurations that differ only by a permutation of identical particles. For the mean energy at fixed N and V , it contributes only a β -independent constant to $\log Z_N$, so it disappears under $-\partial/\partial\beta$. It is not disposable in every question: together with h^{3N} , it is essential for a dimensionless partition function, an extensive entropy, and a consistent chemical potential.

Now set

$$x = \sqrt{\frac{\beta}{2m}} p, \quad dp = \sqrt{\frac{2m}{\beta}} dx. \quad (1.33)$$

The Gaussian integral becomes

$$\int_{-\infty}^{\infty} dp e^{-\beta p^2/(2m)} = \sqrt{\frac{2m}{\beta}} \int_{-\infty}^{\infty} dx e^{-x^2} \quad (1.34)$$

$$= \sqrt{\frac{2\pi m}{\beta}}. \quad (1.35)$$

Substitution into (1.32) gives

$$Z_N = \frac{V^N}{N! h^{3N}} \left(\frac{2\pi m}{\beta} \right)^{3N/2}. \quad (1.36)$$

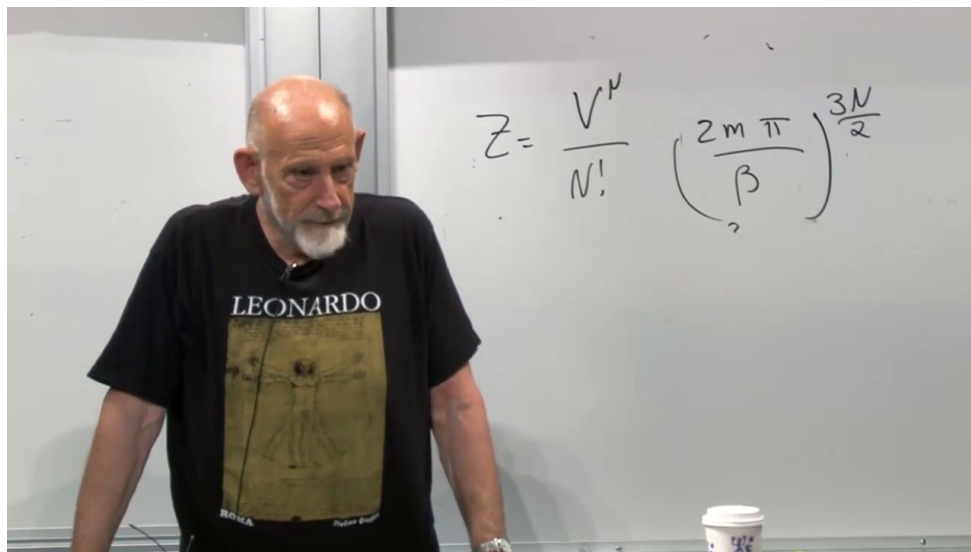


Figure 1.5: The completed ideal-gas partition function on the board. The phase-space cell h^{3N} is restored in (1.36).

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It is useful to package the same result with the thermal de Broglie wavelength

$$\lambda_T = \frac{h}{\sqrt{2\pi m T}} \quad (k_B = 1). \quad (1.37)$$

Then

$$Z_N = \frac{1}{N!} \left(\frac{V}{\lambda_T^3} \right)^N. \quad (1.38)$$

For macroscopic N , Stirling's formula gives

$$N! \simeq N^N e^{-N} \sqrt{2\pi N}. \quad (1.39)$$

Keeping only the leading exponential dependence and defining $\rho = N/V$,

$$\frac{V^N}{N!} \simeq \frac{1}{\sqrt{2\pi N}} \left(\frac{e}{\rho} \right)^N. \quad (1.40)$$

This is the lecture's neat density reorganization. The square-root factor is subleading in $\log Z_N$, but it is displayed here rather than silently dropped.

Question from the audience. If $1/N!$ is independent of β , does it make no difference at all?

Response. It makes no difference to the present mean-energy derivative at fixed N and V , because it adds a β -independent constant to $\log Z_N$. It does matter for absolute entropy and for variations in particle number. The limited derivative statement and the broader state-counting statement should not be confused. ¹⁷

¹⁷Lecture timestamp: 01:27:01–01:27:51.

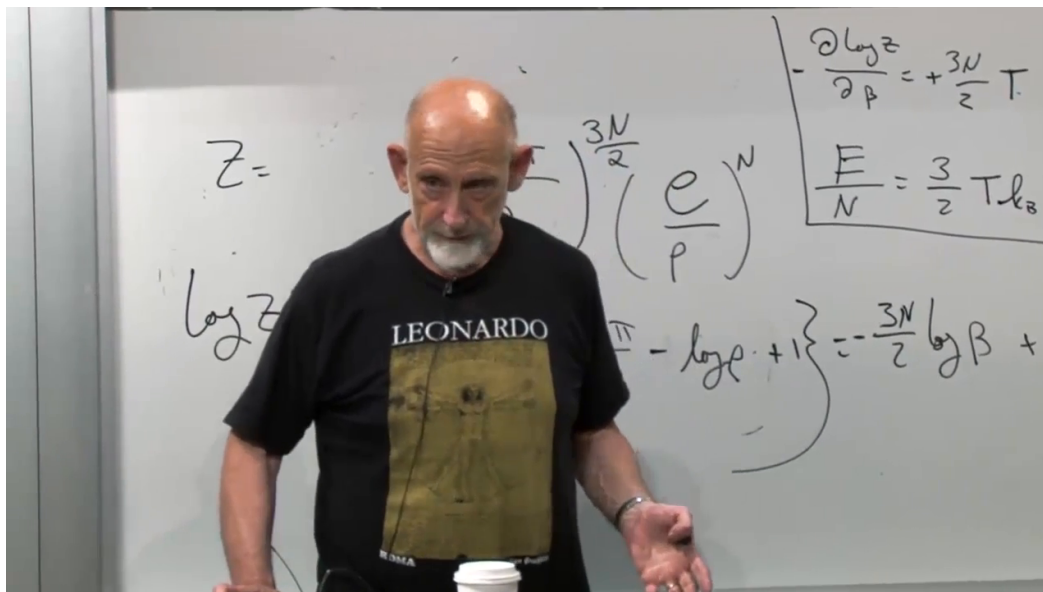


Figure 1.6: Only the $-3N \log \beta/2$ term contributes to the ideal-gas energy, giving $E/N = 3T/2$.

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1.8 Energy and Equipartition

Products are awkward to differentiate; logarithms turn them into sums. From (1.36),

$$\log Z_N = -\frac{3N}{2} \log \beta + \beta\text{-independent terms.} \quad (1.41)$$

Therefore

$$E = -\frac{\partial \log Z_N}{\partial \beta} = \frac{3N}{2\beta} = \frac{3N}{2} T. \quad (1.42)$$

The energy per particle is

$$\frac{E}{N} = \frac{3}{2} T \quad (k_B = 1), \quad (1.43)$$

or, in laboratory units,

$$\frac{E}{N} = \frac{3}{2} k_B T_K. \quad (1.44)$$

Question from the audience. Where does Boltzmann's constant enter the formula for the energy?

Response. The theorist's temperature T is an energy. To use kelvin, replace it by $k_B T_K$, giving $E/N = 3k_B T_K/2$. The small value of k_B converts a macroscopic-looking numerical temperature into the microscopic energy scale of one molecule. ¹⁸

The three comes from the three directions of space. The one-half comes from the square-root dependence of each Gaussian integral on $1/\beta$. Each independent quadratic momentum coordinate contributes

$$\frac{1}{2} T \quad (1.45)$$

¹⁸Lecture timestamp: 01:24:58–01:26:01.

to the mean energy. In d spatial dimensions the same argument gives $E/N = dT/2$. This is the first appearance of equipartition.

The entropy calculation assigned at the end of the lecture follows from $S = \beta E + \log Z_N$. Using Stirling's approximation,

$$\frac{S}{N} \simeq \log \left[\frac{V}{N} \left(\frac{2\pi m T}{h^2} \right)^{3/2} \right] + \frac{5}{2}, \quad (1.46)$$

up to the subleading finite- N correction. Restoring k_B multiplies the right-hand side by k_B and replaces T by $k_B T_K$. The result makes clear why the normalization factors that dropped out of the energy are indispensable for the entropy.

1.9 Putting the Gas in a Gravitational Field

A question about placing the box on a neutron star introduces the gravitational extension. Before confronting a neutron star, consider a uniform field g in an ordinary room. Choose $y = 0$ at the bottom of a box of base area A and height L . The Hamiltonian becomes

$$H = \sum_{a=1}^N \left(\frac{\mathbf{p}_a^2}{2m} + mgy_a \right). \quad (1.47)$$

The horizontal position integrals still give A^N , while each vertical coordinate contributes

$$I_y(L) = \int_0^L dy e^{-\beta mgy} = \frac{1 - e^{-\beta mgL}}{\beta mg}. \quad (1.48)$$

Question from the audience. Should the vertical integral really run to infinity, or only to the top of the box?

Response. For a finite box it must run from 0 to its height L , as in (1.48). Taking the upper limit to infinity is a later simplification that models an atmosphere above a surface rather than a finite room. ¹⁹

The full finite-height partition function is therefore

$$Z_N(L) = \frac{1}{N! h^{3N}} \left[A \frac{1 - e^{-\beta mgL}}{\beta mg} \left(\frac{2\pi m}{\beta} \right)^{3/2} \right]^N. \quad (1.49)$$

The vertical factor depends on β , so it must contribute to the energy. Indeed,

$$\langle U \rangle = N \left[\frac{1}{\beta} - \frac{mgL}{e^{\beta mgL} - 1} \right]. \quad (1.50)$$

This completes the small piece of integration that the lecture first leaves as a nuisance calculation.

For an infinitely high atmosphere, $L \rightarrow \infty$, and

$$I_y(\infty) = \frac{1}{\beta mg}. \quad (1.51)$$

¹⁹Lecture timestamp: 01:30:25–01:31:32.

$$Z = \left(\frac{2m\pi}{\beta} \right)^{\frac{3N}{2}} \left(\frac{1}{\beta mg} \right)^N \frac{A^N}{N!}$$

$$\log Z = -\frac{3N}{2} \log \beta - N \log \beta + \log \frac{A^N}{N!}$$

Figure 1.7: The infinite-atmosphere partition function adds $-N \log \beta$ to $\log Z$, the source of the mean potential energy.

Lecture frame: 01:36:38.

The partition function becomes

$$Z_N^{(\infty)} = \frac{1}{N! h^{3N}} \left[\frac{A}{\beta mg} \left(\frac{2\pi m}{\beta} \right)^{3/2} \right]^N. \quad (1.52)$$

Its β -dependent logarithm is

$$\log Z_N^{(\infty)} = -\frac{3N}{2} \log \beta - N \log \beta + \text{const.} \quad (1.53)$$

The first term gives the kinetic energy $3NT/2$; the second gives the mean gravitational potential energy NT . Hence

$$E_{\text{tot}} = \frac{5N}{2} T \quad (1.54)$$

for this idealized unbounded atmosphere.

Question from the audience. Would a very strong gravitational field require another Lagrange multiplier?

Response. Not merely because the field is strong. Gravity changes the Hamiltonian and therefore the energy levels or phase-space weight. The same energy multiplier β remains. A neutron star introduces additional physical complications, but the first modification is still to put the gravitational potential energy into H . ²⁰

²⁰Lecture timestamp: 01:27:58–01:29:24.

1.10 Three Questions at the Close

Question from the audience. Which equation is the Boltzmann distribution, and what does it assert about equilibrium?

Response. The distribution is

$$p_i = \frac{e^{-\beta E_i}}{Z}.$$

It is the maximum-entropy distribution subject to normalization and a fixed mean energy. The mean energy must be computed from the same probabilities, and normalization must still be enforced even though it looks almost automatic in the final formula. ²¹

The procedure can feel mechanical: write a partition function, take a logarithm, and differentiate. Yet the resulting equations continue to be surprising. Familiarity teaches which simple operation should follow which, but it does not make the subject less curious.

Question from the audience. Can these entropy ideas be applied to a collapsing star or a black hole? How can entropy rise while the matter occupies a smaller region?

Response. Stars are difficult mainly because their material details are complicated. A black hole has the cleaner benchmark: its Bekenstein–Hawking entropy is the limiting gravitational entropy for the given macroscopic constraints. Collapse can increase entropy until that black-hole value is reached; increasing it further requires changing the available energy or size. ²²

Question from the audience. Can the partition function also calculate fluctuations in energy?

Response. Yes. One derivative gives the mean energy, and a second derivative gives its variance:

$$\begin{aligned} \frac{\partial^2 \log Z}{\partial \beta^2} &= \langle E^2 \rangle - \langle E \rangle^2 \\ &= \text{Var}(E). \end{aligned}$$

In energy units this is related to the heat capacity by $\text{Var}(E) = T^2 C_V$. In laboratory units, $\text{Var}(E) = k_B T_K^2 C_V$. This fluctuation relation is the next calculation. ²³

Fluctuations distinguish statistical mechanics from a purely continuous thermodynamic description. Their systematic study, associated especially with Gibbs and Einstein, exposes the finite microscopic constituents behind the smooth macroscopic laws. That is where the next lecture begins.

²¹Lecture timestamp: 01:38:14–01:38:45.

²²Lecture timestamp: 01:39:29–01:40:36.

²³Lecture timestamp: 01:40:37–01:41:53.